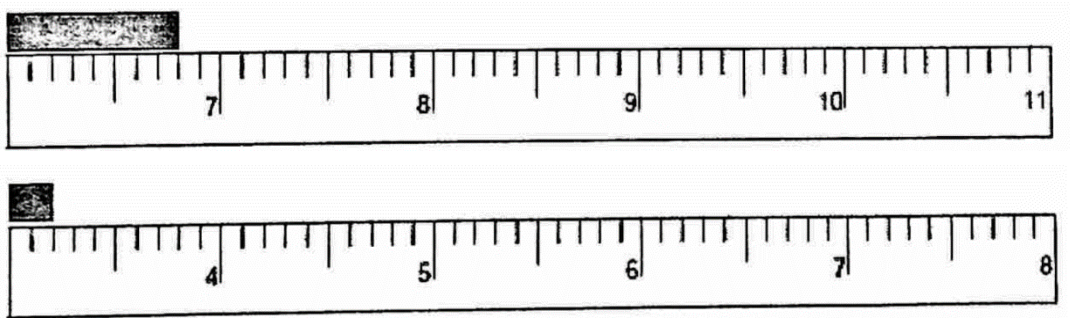
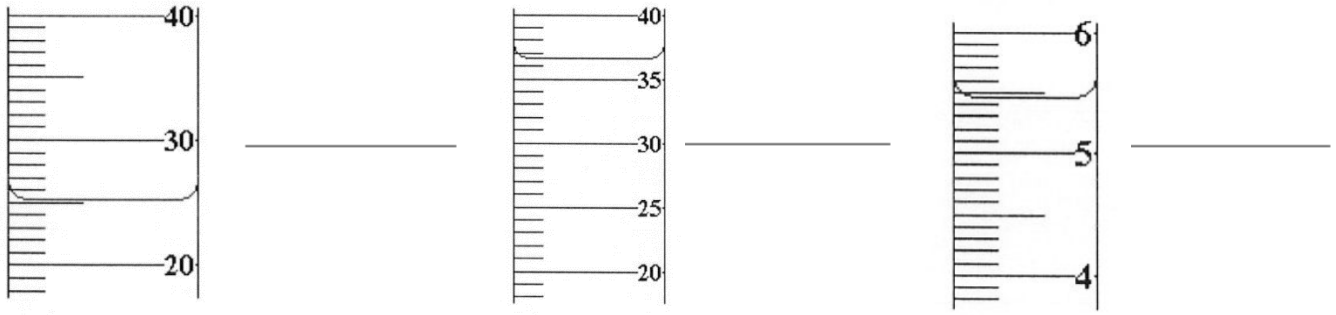
Topic 0.1 – Making Accurate Measurements **Name:** _____

1. What value should be reported in the following measurements?



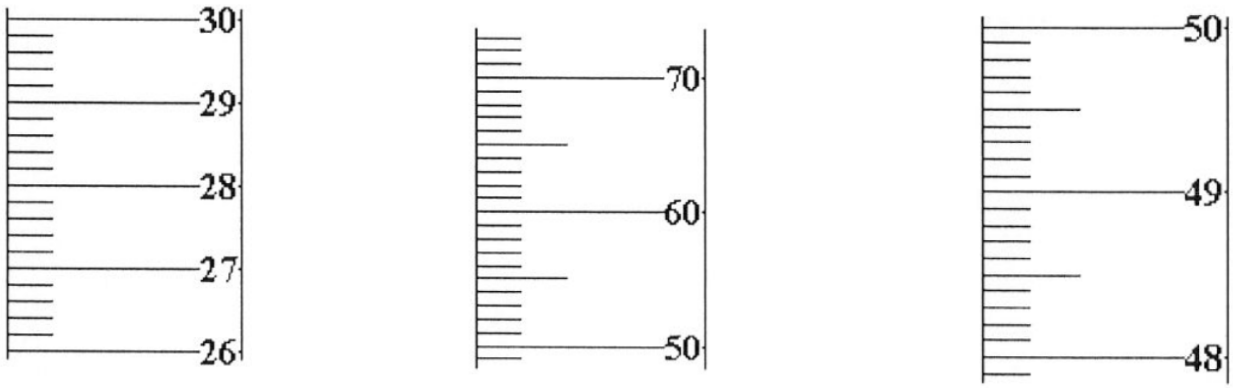
_____

_____



_____ _____ _____

2. Draw the meniscus for the following readings.



27.65 mL 63.8 mL 49.21 mL

Topic 0.2 – Significant Figures and Scientific Notation **Name: _____**

1. How many significant figures are in the following numbers?
   a. 1.050 _____
   b. 1050 _____
   c. 0.01050 _____
   d. 130200 _____
   e. $1.5x10^2$ _____
   f. 500 _____
   g. 100. _____
   h. 0.0100 _____

2. Round 742,396 to four, three, and two significant figures:
   a. Four S.F. _____
   b. Three S.F. _____
   c. Two S.F. _____

3. Round 0.07284 to four, three, and two significant figures:
   a. Four S.F. _____
   b. Three S.F. _____
   c. Two S.F. _____

4. Round 231.45 to four, three, and two significant figures (USE SCIENTIFIC NOTATION)
   a. Four S.F. _____x10—
   b. Three S.F. _____x10—
   c. Two S.F. _____x10—

5. Round 1,350,400 to four, three, and two significant figures (USE SCIENTIFIC NOTATION)
   a. Four S.F. _____x10—
   b. Three S.F. _____x10—
   c. Two S.F. _____x10—

Topic 0.3 – Calculations with Significant Figures

For the following problems, write the answer you get from the calculator and then write the answer rounded to the correct number of significant figures. Don't forget your units!

	Calculator Answer	Answer w/ Sig Figs
a) 3.45 cm + 3.4 cm =		
b) 3.43 cm – 0.056 cm =		
c) 8.4 m + 10 m =		
d) 8.4 m + 10.0 m =		
e) 45 g + 8.345 g =		
f) 5.43 cm x 3.4 cm =		
g) 2.5 in x 3.75 in =		
h) 8.001 m / 2.3 s =		
i) 6.00 g / 0.05 mL =		
j) 5.01 g / 1,002,00 molecules =		

Topic 0.4 – Dimensional Analysis #1

1.00 foot = 12.0 inches	**1.00 inch = 2.54 centimeters**	**60 minutes = 1 hour**
3.00 feet = 1.00 yard	**1.00 kilometer = 0.62 miles**	**24 hour = 1 day**
5280 feet = 1.00 mile	**60 seconds = 1 minute**	**365 days = 1 year**
1.00 pound = 16.0 ounces	**1.00 gallon = 3.79 liters**	

ALL work should be shown on the problems below, even if you can do it in your head. You do not get credit on a test for doing the work in your head. Round ALL answers to the correct number of significant figures.

1. How many inches are in 36.35 centimeters?

2. A football field is 100.0 yards from goal line to goal line. Calculate the length of the football field in miles.

3. The moon is roughly 250,000 miles away. Calculate the distance in feet.

4. How many centimeters are in 2.15 miles?

5. A car is traveling at a speed of 35.0 miles/hour. Calculate the speed of the car in units of feet per minute.

6. The cost of gold is $2340 per ounce. How many pounds of gold can someone buy with $1500?

Topic 0.5 – Density

1. What does density measure?

2. A student is comparing 2 blocks of pure aluminum. Block A is much larger than Block B. Do you expect block A's density to be lower, higher, or the same as Block B? Explain your answer.

3. Mercury metal is poured into a graduated cylinder and the volume is measured to be 22.5 mL. The mass of that sample of mercury is 305.4 g. From this information, calculate the density of the mercury sample.

4. A rectangular block of copper metal weighs 1896 g. The dimensions of the block are 8.4 cm by 5.5 cm by 4.6 cm. From this data, calculate the density of the copper metal. Consider the unit needed for volume.

5. Find the mass of 250.0 mL of benzene. The density of benzene is 0.8765 g/mL.

6. 28.5 g of iron is added to a graduated cylinder containing 45.50 mL of water. The water level rises to the 49.10 mL mark. From this information, calculate the density of the iron.

7. What is the volume of a liquid, in mL, that contains 25.0 grams of the substance with a density 1.20 g/mL?